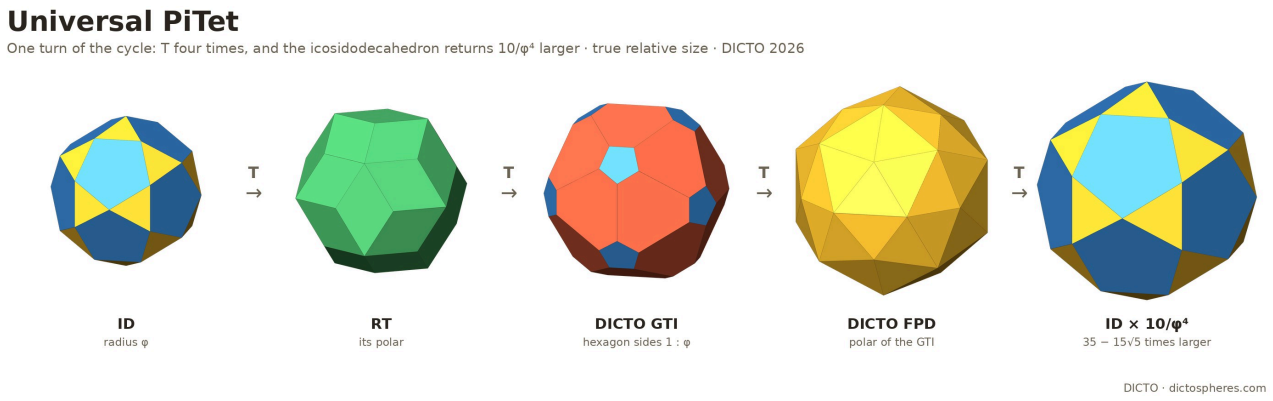
Universal PiTet

The DICTO PiTet Proposition: an endless, self-polar four-solid cycle grown from the Cleo star, ID → RT → DICTO GTI → DICTO FPD → ID × 10/φ⁴, with exact shell pieces, five-sided Vajras and hexagonal Cleos

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Version 3 · 20 October 2026 (new figures, drawn from the exact solids and pieces; prior art for the rule T added in §2 and §6: T is the three-dimensional form of the classical antipedal construction; Version 2 corrected one sentence in §5 and one word in the abstract; licence CC BY-SA 4.0) · DICTO DOI: 10.5281/zenodo.23463305 · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra (research branch, studies/2026-10-19-five-fold-octet) · Interactive: dictospheres.com → Kaleidohedra → DICTO → Universal PiTet



Abstract. DICTO specified a five-fold version of the octet: spherical shells grown outward from the Cleo star, the same pattern each time with different solids, triangles and pyramids staying constant (hence the name PiTet: Pi for the spherical progression, Tet for the triangles). We give a rule that does this and prove, in exact arithmetic in $\mathbb{Q}(\sqrt{5})$, what it produces. Let T take a convex solid with corners v to the solid $\{x : v\cdot x \le v\cdot v$ for every corner $v\}$: each new face passes through a corner of the last solid at right angles to it. Starting from the icosidodecahedron (ID), T gives the rhombic triacontahedron (RT), then the **DICTO GTI** (Golden Truncated Icosahedron: hexagon sides 1 : φ), then the **DICTO FPD** (Flat Pentakis Dodecahedron), and then the ID again, exactly $\mathbf{10/\varphi^4 = 35 - 15\sqrt{5}}$ times larger; the cycle therefore repeats without end. Each solid lies in the next, each shell is filled by congruent pieces of one or two kinds (exact volumes), and the whole chain is its own polar about the GTI's circumsphere. Below the cycle, the Cleo star (82 pieces) and 12 twisted frusta (Monkey Staff B) close the ID; the pieces on one five-fold axis join into a convex five-sided Vajra, and the DICTO Ogdoecontahedron fits between the ID and the RT with 20 pure hexagonal Cleos. Every claim is checked exactly.

1. The idea

Nested polyhedra with one centre are old: Kepler's *Mysterium cosmographicum* [14] set the Platonic solids between concentric spheres, and Pacioli and Leonardo drew them as skeletons [15]. They are also physical: the atomic clusters of icosahedral quasicrystals are concentric shells on the same solids, a dodecahedron inside an icosahedron and icosidodecahedron, or an icosahedron inside a dodecahedron [16], and Mackay's dense icosahedral packing of equal spheres grows in icosahedral shells [17]. The octet truss fills flat space with tetrahedra and

octahedra [18]. Curved space can do it with tetrahedra alone (the 600-cell [4, 7]), and seen from one vertex the 600-cell's shells begin icosahedron, dodecahedron, icosahedron, icosidodecahedron [19]. DICTO's idea was to build a five-fold, spherical version in our own space: start from the Cleo star (the icosahedron as 20 Cleo-Icos with 20 more turned out as spikes, DICTO DOI: 10.5281/zenodo.23378070), and grow outward shell by shell, the same pattern each time with different solids, with multi-sided Vajras in the place of the octet's octahedra and kinds of CLEO and FUJI among the pieces. He observed that triangles are the true constant: in every shell built here, every face where two pieces meet is a triangle, and pentagons, hexagons and rhombi appear only on the shells' surfaces.

2. The rule and the DICTO PiTet Proposition

The rule. For a convex solid S with corners v, let $T(S) = \{x : v \cdot x \le v \cdot v \text{ for all corners } v \text{ of } S\}$. Each face of T(S) passes through a corner of S, at right angles to the line from the centre to that corner, so the corners of S touch the faces of T(S) ("points meeting", DICTO). When all the corners of S lie on one sphere, T is ordinary polar reciprocation about that sphere; when they lie at several radii it is not, and that difference is what drives the cycle. The rule needs no square roots, so everything below stays in $\mathbb{Q}(\sqrt{5})$. In the plane the same construction about a point is classical: the *antipedal* polygon, whose sides pass through the vertices at right angles to the lines from the point; T is its three-dimensional form about the centre (see §6) [26, 27].

The DICTO PiTet Proposition. Let ID be the icosidodecahedron of circumradius φ. Then (i) T(ID) is the rhombic triacontahedron RT, T(RT) is the DICTO GTI, T(GTI) is the DICTO FPD, and $T(FPD) = (10/\varphi^4)\cdot ID$ exactly; hence $T^{4n}(ID) = (10/\varphi^4)^n \cdot ID$ for every n, since T commutes with scaling. (ii) Each solid lies inside the next. (iii) Each shell between consecutive solids is filled exactly by one piece per face of the inner solid (the cone from the centre over that face, cut by the outer solid), the pieces falling into one or two congruence classes (Table 2). (iv) The chain is its own polar: reciprocation about the sphere of radius^2 $15 - 5\sqrt{5} = 10/\varphi^2$ (the GTI's circumsphere) swaps GTI ↔ FPD, RT ↔ $ID\cdot 10/\varphi^4$ and ID ↔ $RT\cdot 10/\varphi^4$.

Proof method. All coordinates are pairs of rationals $a + b\sqrt{5}$ with exact signs. Floating point is used only to guess which three faces meet at a corner; every guess is then proved exactly: each corner solves its faces and lies inside every face; every face holds at least three corners with all others inside it; every edge lies in exactly two faces and $V - E + F = 2$, so the corner list is complete. Congruence of pieces is proved by the 60 exact rotations of the icosahedral group; volumes are exact; the shells' sums equal their gaps with remainder exactly 0. The scripts (exact_cycle.py, exact_shells.py, exact_polar.py) run in seconds.

3. The four stations

Station	Faces	Corners (radius)	Note
ID	20 triangles, 12 pentagons	30 (φ)	start of each turn
RT	30 rhombi	12 ($2R_I \approx 1.902$), 20 ($\sqrt{3}$)	polar of the ID
DICTO GTI	12 regular pentagons, 20 hexagons, sides 1 : φ	60 ($\sqrt{10}/\varphi \approx 1.954$)	the icosahedron cut at 1/(2+φ) of each edge, not 1/3; all corners on one sphere
DICTO FPD	60 isosceles triangles	12 (≈ 2.008), 20 (≈ 2.205)	polar of the GTI; flatter pyramids than the Catalan pentakis dodecahedron
$ID\cdot 10/\varphi^4$	the ID again		next turn

Table 1. The stations of one turn (radii in the frame where the icosahedron has edge 1 and R_I = its circumradius). The cycle alternates polar steps (from the ID and the GTI, whose corners lie on one sphere) and non-polar steps (from the RT and the FPD); the non-polar steps join the two classical dual pairs ID–RT and GTI–FPD into one cycle and fix the gain $10/\varphi^4$.

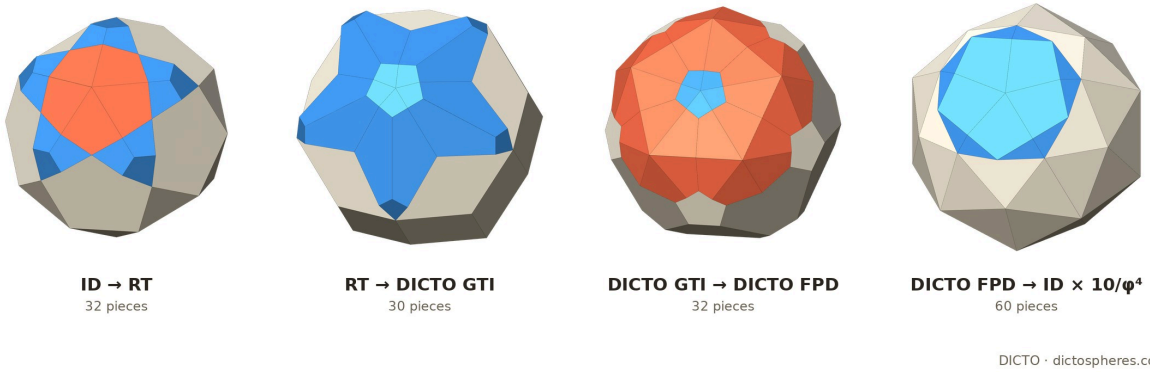


Figure 1. The four shells of one turn, each filled exactly by one piece per face of the inner solid (pale): ID → RT, RT → GTI, GTI → FPD and FPD → ID·$10/\varphi^4$, with the pieces on the faces round one five-fold axis, one colour per kind.

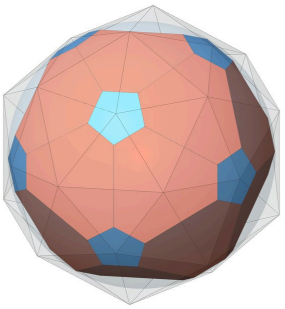


Figure 2. The mirror of Proposition (iv): the DICTO GTI inside the DICTO FPD (glass), with the sphere through the GTI's corners (radius² $15 - 5\sqrt{5}$). Reciprocation about this sphere swaps the two, and reflects the whole chain.

4. The shell pieces (exact)

Shell	Pieces	Exact volume of each	Gap (exact)
ID → RT	12 on pentagons, 20 on triangles	$5/12$; $3/8 - 17\sqrt{5}/120$	$25/2 - 17\sqrt{5}/6$
RT → GTI	30 on rhombi	$(3 - \sqrt{5})/4 = 1/(2\varphi^2)$	$45/2 - 15\sqrt{5}/2$
GTI → FPD	12 on pentagons, 20 on hexagons	$(67475 - 30175\sqrt{5})/72 \approx 0.0229$; $(5681 - 2534\sqrt{5})/36 \approx 0.4112$	$(259235 - 115865\sqrt{5})/18$
FPD → ID·$10/\varphi^4$	60 on triangles	$(77875 - 34825\sqrt{5})/27 \approx 0.1457$	$(1557500 - 696500\sqrt{5})/9$

Table 2. One piece per face of the inner solid; each kind is one symmetry orbit, so its pieces are congruent; each next turn repeats them $10/\varphi^4$ larger. In every shell the walls between neighbouring pieces are triangles.

5. From the Cleo star to the cycle (exact)

Stage 1, the Cleo star. The icosahedron of edge 1 as 20 Cleo-Icos, 20 spikes, 12 inward pentagonal pyramids and 30 valley tetrahedra: 82 pieces filling the dodecahedron D′ of

circumradius $\varphi^2/\sqrt{3}$ exactly, with no overlap.

Stage 2, Monkey Staff B. The tightest ID round D′ has every corner of D′ exactly at the centre of one of its triangles. The gap is filled by 12 congruent twisted frusta, DICTO's **Monkey Staff B** (the D′ pentagon below, the ID's pentagon above turned 36°, ten triangles round the side), each of volume $(145 + 37\sqrt{5})/648$.

Stage 3 and the five-sided Vajra. The ID → RT shell can also be cut into 12 five-sided and 20 three-sided pyramids and 60 valley tetrahedra (92 pieces, exact, no overlap). (Version 1 called the three-sided pyramid DICTO's CLEO. A corner-by-corner shape test shows it is not: it has a lower point, about 2.17 times its base radius high against CLEO's $\varphi^2 \approx 2.618$.) On each five-fold axis the Cleo star's inward pyramid, Monkey Staff B and the outward five-sided pyramid join into one convex double-pointed piece, a **five-sided Vajra** of volume $55/108 + 5\sqrt{5}/36$, reaching from the icosahedron's corner to the RT's.

Hexagonal Cleos. The DICTO Ogdoecontahedron (OG; DICTO DOI: 10.5281/zenodo.23404688) fits exactly between the ID and the RT: at this size its corners are the ID's 30 corners, the 60 points 2c − v (v a corner of an ID triangle, c its centre) and 12 apexes, and its 20 regular hexagons lie in the ID's triangle planes, each centred on a corner of D′. ID → OG is filled by 12 congruent pieces; OG → RT by **20 pure hexagonal Cleos** (regular hexagon base, apex at the RT's three-fold corner, each $1/4 - \sqrt{5}/12$), 60 kite pyramids from the RT's five-fold corners (each $1/76 + \sqrt{5}/684$) and 120 tetrahedra (each $1/24 - \sqrt{5}/72$), all exact, with no overlap.

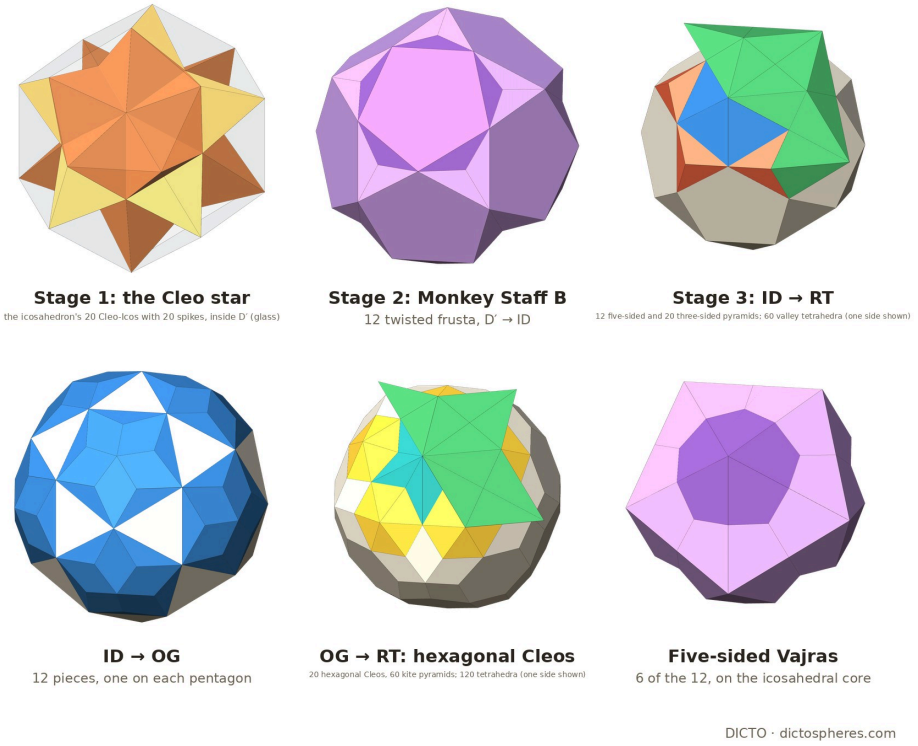


Figure 3. From the Cleo star to the cycle. Top: stage 1, the Cleo star in D′ (glass); stage 2, Monkey Staff B on D′; stage 3, the ID → RT pyramids with the valley tetrahedra on one side. Bottom: the 12 ID → OG pieces; OG → RT, the hexagonal Cleos (yellow) and kite pyramids (teal) with the tetrahedra on one side; six of the twelve five-sided Vajras on the icosahedron.

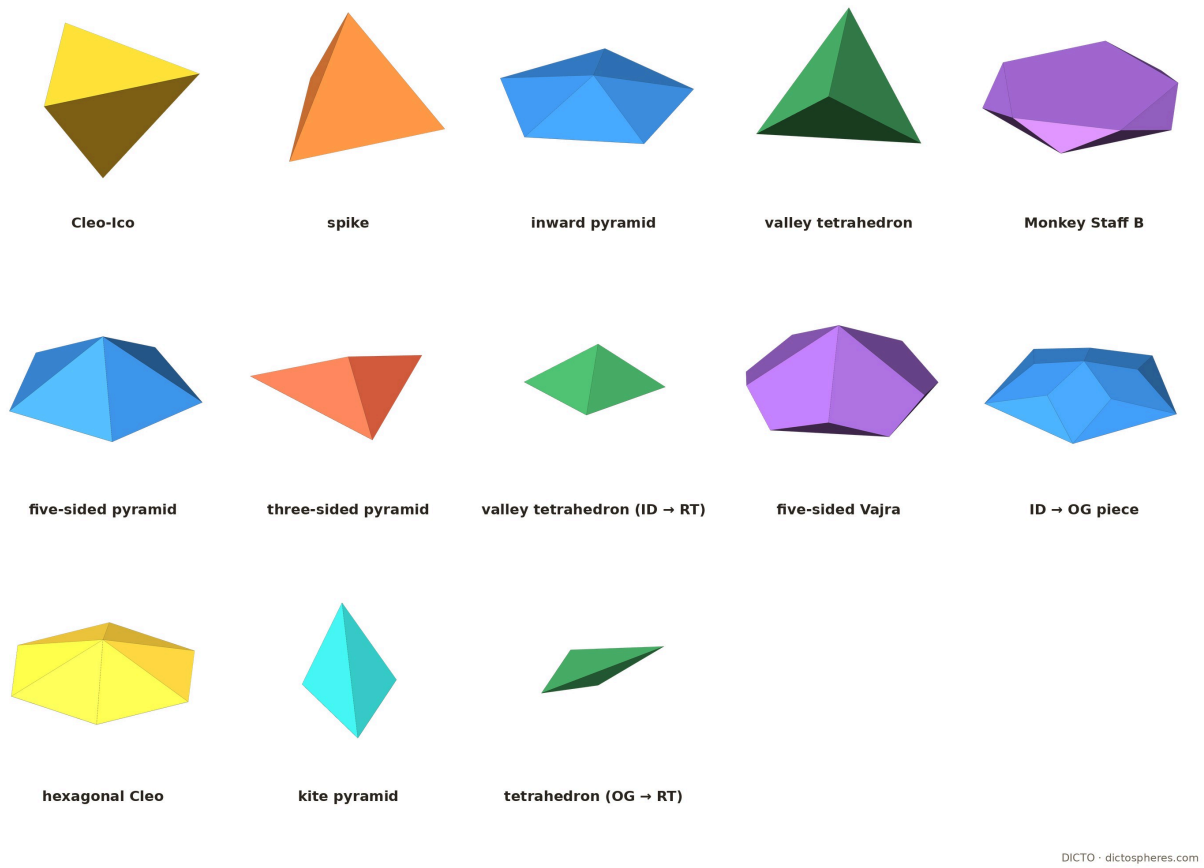


Figure 4. One piece of each kind, each to its own scale, coloured as in Figure 3.

6. Prior art and what is claimed

Endless shells round a pyramid core. The closest prior art we found is Bergman's continuous spherical truss [1]: an icosahedral nucleus of twenty tetrahedral pyramids surrounded by an endless sequence of space-filling icosahedral-spherical tetrahedron–octahedron layers of equal thickness, every layer of the same kind. The idea of a pyramid core with endless octet-like spherical shells is therefore not new, and we do not claim it. Lalvani [6] built icosahedral space structures from polygonal members along the two-fold axes, periodic or not, without a centre.

Central icosahedral fillings. Kramer [2] filled space round a centre with icosahedral symmetry using seven convex cells that pack into a dodecahedron and into copies of themselves τ^3 larger [20]; Mosseri and Sadoc reduced the cells to four [20, 24], and two of those are DICTO's AXE and CLEO [11, 21]. That growth is a golden self-similar inflation; the UP gain $10/\varphi^4 \approx 1.459$ is not a power of φ, and UP's shells are four different solids, not copies of one.

Duals, reciprocation and the stations. The ID and RT are a classical dual pair [4, 5]; Hart's canonical polyhedra make any dual pair meet at a common midsphere [3], and plain reciprocation has period 2. Truncated icosahedra with two edge lengths arise as orbits of the icosahedral group [19, 22]; the DICTO GTI is the member whose hexagon sides are in the ratio $1 : \varphi$, and the DICTO FPD is its polar, a pentakis dodecahedron flatter than the Catalan one. What is claimed is these golden proportions as forced by the rule; their faces are not regular, but they are convex, planar, fully icosahedral and exact. The RT's stellations [23] and the Kepler–Poinsot solids [4] belong to the taller version of §7.

The rule T (added in Version 3). T is the three-dimensional form of the antipedal construction of plane geometry, and its reverse, whose corners are the feet of the perpendiculars from the centre to the faces, is the “face pedal polyhedron” that Smarandache posed as an open problem in 2010 [26]; iterated pedal and antipedal triangles are studied in [27]. A later search for a companion paper found these; Versions 1 and 2 reported no source for T. We found no study of T for polyhedra, of its iteration, or of the cycle below.

Not found. In the sources checked (the full text of the 87 library sources listed in the appendix, and web searches on 19 October 2026 for iterated reciprocation, nested icosahedral shells, and the four stations in sequence) we did not find the rule T applied to polyhedra, the four-step cycle with gain $10/\varphi^4$, its self-polarity about the GTI's circumsphere, the shell pieces, Monkey Staff B, the five-sided Vajra or the hexagonal Cleos in OG hexagons.

7. Further work

The same rule run outward from the OG gives the DICTO KRE, PHE, TPP and DPHP, which nest only as far as the TPP (to be reported separately). A taller version with true Kepler–Poinsot spikes (the great and small stellated dodecahedra alternating, gain φ^3 a turn) is being studied, following DICTO's "think stellation, think Gaudí". Twelve OGs packed freely round one show no symmetry at all (float search), consistent with the geometrical frustration of icosahedral order in flat space [7, 25] that the shells sidestep.

Data and reproducibility

Exact scripts: exact_cycle.py (the cycle), exact_shells.py (the shell pieces), exact_inner.py (Cleo star, Monkey Staff B, fine cut, OG stage, Vajra), exact_polar.py (self-polarity); float exploration: progression.py, hexstage.py, stage4.py, stage5.py, stepper.py; results in data/*.json. Figures (Version 3): up_figs.py (float copies of the exact solids and pieces; it also checks the 60 kite pyramids exactly), shots.mjs, up_compose.py. The interactive world (shells, Vajras, hexagonal Cleos, and the cycle film peeling station by station without end) is in the Kaleidohedra app.

References

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5. M. J. Wenninger, Dual Models, Cambridge University Press (1983).
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9. DICTO, The DICTO Ogdoecontahedron, DICTO DOI: 10.5281/zenodo.23404688.
10. DICTO, Spheres of Dharma, DICTO DOI: 10.5281/zenodo.23310833.
11. DICTO, AXE, CLEO and FUJI: the Gap Pieces of the Icosahedron, DICTO DOI: 10.5281/zenodo.23351291.
12. DICTO, Vajrayana, DICTO DOI: 10.5281/zenodo.23429911.
13. DICTO, The DICTO Ogdoecontahedron on the Octet Truss, DICTO DOI: 10.5281/zenodo.23457016.
14. J. Kepler, Mysterium cosmographicum (1596).

15. L. Pacioli, Divina proportione, illustrations by Leonardo da Vinci (1509).
16. N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai, Cluster packing geometry for Al-based F-type icosahedral alloys, arXiv:1301.6823 (2013).
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20. P. Kramer, Gateways towards quasicrystals, arXiv:1101.0061 (2011).
21. N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi, Dodecahedral structures with Mosseri–Sadoc tiles (2020).
22. N. Al-Siyabi, N. O. Koca, M. Koca, Icosahedral polyhedra from the D6 lattice and Danzer's ABCK tiling (2020).
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Appendix. The reference library searched

DICTO's reference library (private copies, with public citations in the Kaleidohedra repository's docs/SOURCES.md). The full text of the 87 sources below was searched on 19 October 2026 for this paper's subjects: polar reciprocation and duals; nested or concentric polyhedral shells; the icosidodecahedron, rhombic triacontahedron, truncated icosahedra and pentakis dodecahedra; the octet truss; the 600-cell; Mackay and quasicrystal clusters; stellations and the Kepler–Poinsot solids. Bergman [1] was added to the library for this paper.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
E. S. Fedorov	1885	Elements of the Theory of Figures (Nachala ucheniya o figurah)
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
R. Buckminster Fuller	1954	Building construction (geodesic dome)
R. Buckminster Fuller	1961	Synergetic building construction (octet truss)
R. Buckminster Fuller	1962	Tensile-integrity structures
Kenneth D. Snelson	1965	Continuous tension, discontinuous compression structures
Peter J. Pearce	1971	Synthesized natural geometric structures
Stephen C. Baer	1973	Structural system
John P. Hogan	1976	Homohedral construction employing icosahedron
Peter J. Pearce	1976	Minimum inventory maximum diversity building system
Haresh Lalvani	1988	Building structures based on polygonal members and icosahedral symmetry
Helmut Bergman	1988	Continuous spherical truss construction (US 4,719,726)
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
H. Lalvani	1995	Periodic space structures composed of two nodal polyhedra for design applications (US 5,448,868)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
H. Lalvani	1997	Building systems with non-regular polyhedra based on subdivisions of zonohedra (US 5,623,790)
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra

M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral Polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian et al.	2020	Bi-axial woven tiles: interlocking space-filling shapes based on symmetries of bi-axial weaving patterns
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